



Hospital AI Chatbot

using Python and Natural Language Processing (NLTK)

PROJECT REPORT

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01 Introduction

Nowadays, hospitals receive many repetitive questions from patients such as doctor availability, consultation fees, emergency services, laboratory timings, pharmacy information, and appointment guidance. Answering these questions manually takes time and increases the workload of reception staff.

To solve this problem, I developed a **Hospital AI Chatbot** using Python and Natural Language Processing (NLTK). The chatbot understands predefined user queries and provides quick and relevant responses based on a custom dataset.

02 Problem Statement

Patients often need basic hospital information before visiting a hospital. Reception staff spend a significant amount of time answering the same questions repeatedly. Some common questions include:

- Which doctor should I consult?
- What are the doctor's timings?
- What is the consultation fee?
- Where is the laboratory?
- Is the pharmacy open?
- How can I book an appointment?
- What are the visiting hours?

Providing this information manually can be time-consuming and inefficient.

03 Proposed Solution

To solve this problem, a Hospital AI Chatbot was developed. The chatbot reads user questions, identifies the user's intent using Natural Language Processing techniques, and returns the most appropriate response from the dataset.

The chatbot provides information related to:

- Doctors
- Departments
- Emergency Services
- Ambulance
- Laboratory
- Pharmacy
- ICU
- Blood Bank
- Appointment Guidance

- Consultation Fees
- Visiting Hours
- Health Tips

04 Objectives

The objectives of this project are:

- To build a hospital information chatbot.
- To reduce the workload of reception staff.
- To provide quick responses to patients.
- To demonstrate the use of Natural Language Processing.
- To improve user experience through automated responses.

05 Technologies Used

The following technologies were used in this project:

- Python 3.11
- NLTK (Natural Language Toolkit)
- NumPy
- JSON
- Pickle
- Visual Studio Code

06 Project Workflow

The chatbot works in the following steps:

- 1 The user enters a question.
- 2 The chatbot tokenizes the sentence using NLTK.
- 3 A vocabulary is created from the training dataset.
- 4 The chatbot compares the input with the trained vocabulary.
- 5 The matching intent is identified.
- 6 A suitable response is selected from the dataset.
- 7 The chatbot displays the response to the user.

07 Dataset

A custom dataset named **intents.json** was created for the chatbot. The dataset contains different hospital-related intents such as:

- Greetings
- Doctor Information
- Departments
- Appointments
- Emergency Services
- Ambulance
- Laboratory
- Pharmacy
- ICU
- Blood Bank
- Consultation Fees
- Visiting Hours
- Health Tips
- First Aid
- Common Symptoms

Each intent contains:

- Tag
- Training Patterns
- Multiple Responses

08 Model

The chatbot generates two model files after preprocessing and training:

- words.pkl
- classes.pkl

These files store the chatbot vocabulary and intent classes, allowing the chatbot to recognize user queries efficiently.

09 Project Structure

```
Hospital-AI-Chatbot
■
■■■ data
■   ■■■ intents.json
■
■■■ model
■   ■■■ words.pkl
■   ■■■ classes.pkl
■
■■■ src
■   ■■■ chatbot.py
■   ■■■ inf.py
■   ■■■ train.py
■
■■■ requirements.txt
■■■ README.md
```

10 Expected Output

Example 1

User	Hello
Bot	Hello! Welcome to CityCare Hospital. How may I assist you today?

Example 2

User	I have fever
Bot	Please consult Dr. Ahmed Raza (General Physician). Timing: 9:00 AM – 2:00 PM Consultation Fee: PKR 2,000

Example 3

User	Need Cardiologist
Bot	Dr. Bilal Siddiqui Timing: 6:00 PM – 9:00 PM Consultation Fee: PKR 3,500

11 Advantages

- Easy to use
- Fast response
- Lightweight application
- User-friendly interface
- Easily customizable dataset
- Reduces repetitive work
- Works offline

12 Limitations

- The chatbot only answers questions available in the dataset.
- It cannot learn new information automatically.

- It does not connect to a hospital database.
- It does not support voice interaction.

13 Future Improvements

The chatbot can be improved by adding:

- Web-based interface
- Mobile application
- Voice support
- Database integration
- Online appointment booking
- Real-time hospital data
- Multilingual support
- AI-powered language models

14 Conclusion

The Hospital AI Chatbot successfully provides quick and accurate responses to common hospital-related queries using Natural Language Processing. The project demonstrates how AI techniques can be applied to automate repetitive tasks and improve communication between hospitals and patients. It is simple, efficient, easy to maintain, and can be further enhanced with additional AI technologies in the future.